

Engineering: Local Development Environment Setup

Version: 2026.4

Department: Engineering

Classification: Internal / Engineering Only

Type: Order of Operations

1. Prerequisites

Before beginning the setup, ensure your machine meets the following requirements:

- **Hardware:** Apple Silicon (M1/M2/M3) preferred.
- **OS:** macOS 14.0 (Sonoma) or higher.
- **Package Manager:** [Homebrew](#) installed.

2. Core Toolchain Installation

Run the following commands in your terminal to install the necessary languages and runtimes:

```
# Update Homebrew
brew update
```

```
# Install Node.js (LTS) and pnpm
brew install node
npm install -g pnpm
```

```
# Install Docker Desktop (Required for local Supabase)
brew install --cask docker
```

3. Repository Setup

1. Clone the Monorepo:

```
git clone
[https://github.com/company/core-platform.git](https://github.com/company/core-platfor
m.git)
cd core-platform
```

2. Install Dependencies:

```
pnpm install
```

4. Environment Configuration

We use a centralized `.env.local` for development.

1. Copy the template:
`cp .env.example .env.local`
2. **Request API Keys:** Reach out to your onboarding buddy or ping `#eng-ops` on Slack to receive your development keys for:
 - `OPENROUTER_API_KEY`
 - `GROQ_API_KEY`
 - `SUPABASE_SERVICE_ROLE_KEY`

5. Database & Backend (Supabase)

We use the Supabase CLI to manage local development databases.

1. **Login to Supabase:**
`npx supabase login`
2. **Start Local Environment:**
`npx supabase start`

This will pull Docker images for Postgres, GoTrue, PostgREST, and Realtime.

6. Running the Application

To start the development server with hot-reloading:

```
pnpm dev
```

The application will be accessible at `http://localhost:3000`.

7. Troubleshooting Common Issues

- **Docker Context Error:** Ensure Docker Desktop is running and you have assigned at least 4GB of RAM to the Docker engine.
- **Port 5432 Conflict:** If you have a native Postgres installation running, stop it using `brew services stop postgresql` before running `supabase start`.
- **Auth Redirects Failing:** Check that `SITE_URL` in `.env.local` is set to `http://localhost:3000`.

Next Steps: Review the `docs/architecture.md` file in the repository to understand our service

mesh and RLS implementation.